

Fig. 5: Program output screenshot of fibre-optic communication link

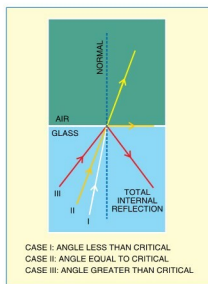


Fig. 6: Program output screenshot of total internal reflection with rays of incidence at different angles

degraded light signal.

Receiver. The receiver includes a fibre-to-light detector interface (light coupler), a photo-detector and a current-to-voltage converter. The fibre-to-light detector is also a mechanical coupler. Its function is to couple as much light as possible from the fibre cable into the light detector.

The light detector is generally a PIN diode, avalanche photo-diode or phototransistor. All the three devices convert light into current. Consequently, a current-to-voltage converter is required to produce an

output voltage proportional to the original source information. Analogue or digital interfaces are electrical interfaces that match impedances and signal levels between the information source and the destination to the input and output circuitry of the optical system.

Total internal reflection

Index of refraction. The amount of bending or refraction that occurs at the interface of two materials of different densities depends upon the refractive indices of the two materials. Thus the index of refraction is the way of measuring the speed of light in a material. It is simply the ratio of the velocity of propagation of light in free space (c) to the velocity of propagation of a light ray in a given material (v).

Refractive index (n) = c/v

Total internal reflection. When light passes from a medium with one index of refraction (say, m_1) to another medium with a lower index of refraction (say, m_2), it bends or refracts away from an imaginary line perpendicular to the surface (normal line). As the angle of the beam through m_1 becomes greater with respect to the normal line, the refracted light through m_2 bends further away from the line.

At one particular angle (critical

angle), the refracted light doesn't go into m_2 , but instead travels along the surface between the two media (sine [critical angle] = n_2/n_1 , where n_1 and n_2 are the indices of refraction ($n_1 > n_2$)). If the beam through m_1 is greater than the critical angle, the refracted beam is reflected entirely back into m_1 (total internal reflection), even though m_2 may be transparent.

The light must strike the boundary at an angle greater than critical angle. The light beam that enters the fibre within a certain range of angles can travel down the fibre without leaking out. This range of angles is called the acceptance cone of the fibre.

However, some of the light signal degrades within the fibre, mostly due to impurities in the glass. The extent to which the signal degrades depends upon the purity of the glass and the wavelength of the transmitted light.

Types of optical fibres

Optical fibres are available in two types: single-mode and multi-mode.

Single-mode fibres have small cores (about 9 microns in diameter) and transmit infrared laser light (having wavelength of 1300 to 1550 nanometres).

Multi-mode fibres have larger cores (about 62.5 microns in diameter) and transmit infrared light (having wavelength of 850 to 1300 nm) from light-emitting diodes (LEDs).

Some optical fibres can be made from plastic. These fibres have a large core (with 1mm diameter) and transmit visible red light (having wavelength of 650 nm) from LEDs.

Software program

The above fibre-optical concepts have been taken into consideration while developing this project. The program is written in 'C' language and compiled using Turbo C IDE. Turbo C has a good collection of graphics libraries. The program uses various 'C' graphics files such as egavga.bgi, scri.chr, trip.chr and tscr.chr. When you run the executable file, following output